

GRB Afterglow Detection Rates: Implementation Reference

Model equations as implemented in `grb.detections`

Summary

The code computes the expected annual detection rate R_{det} [yr^{-1}] of gamma-ray burst (GRB) optical afterglows as a function of survey strategy parameters ($N_{\text{exp}}, t_{\text{cad}}$). The calculation combines a Blandford–McKee ISM afterglow model with a Euclidean population of GRBs and a background-limited survey instrument.

The central insight is that detectability is governed by *two competing horizons*. The **Euclidean horizon** q_{Euc} is the maximum off-axis viewing angle at which a burst at the calibration distance D_{Euc} is brighter than the survey flux limit. The **cadence horizon** q_i is the maximum off-axis angle whose afterglow peak arrives early enough to be caught within i cadence intervals. Comparing these two angles against three characteristic afterglow angles ($q_{\text{dec}}, q_j, q_{\text{nr}}$) divides the strategy space into seven piecewise regimes A1–A7, each with its own analytic rate formula. The surface $\log_{10} R_{\text{det}}(N_{\text{exp}}, t_{\text{cad}})$ is computed on a dense grid and displayed interactively. An optional luminosity function (Section 9) describes the event-rate density over all positive spectral luminosities $L \equiv L_{\nu}(t_{\text{dec}})$. Its two parameters are the differential normalization $\mathcal{A}$ and slope α . Luminosity is integrated analytically in both rate modes.

1 Parameters

Symbol	Default	First value	Unit	Meaning
<i>Physical model</i> (<code>AfterglowPhysicalParams</code>)				
p	2.2	2.5*	–	Electron energy distribution index
$E_{k,\text{iso}}$	10^{53}	(unchanged)	erg	Isotropic kinetic energy
n_0	1.0	(unchanged)	cm^{-3}	ISM ambient number density
θ_j	0.1	(unchanged)	rad	Jet half-opening angle
Γ_0	$10^{2.5}$	(unchanged)	–	Initial bulk Lorentz factor
ν	5×10^{14}	(unchanged)	Hz	Observing frequency (optical V-band)
z_{Euc}	2.0	(unchanged)	–	Alt. cosmology redshift (reserved, unused by default)
H_0	70	(unchanged)	km/s/Mpc	Alt. cosmology Hubble constant (reserved, unused)
Ω_m	0.3	(unchanged)	–	Alt. cosmology matter density (reserved, unused)
<i>Microphysics</i> (<code>MicrophysicsParams</code>)				
ε_e	0.1	(unchanged)	–	Electron energy fraction
ε_B	10^{-4}	0.01*	–	Magnetic field energy fraction
z	0.0	(unchanged)	–	Redshift factor input (reserved, unused by default)
<code>include_redshift_factors</code>	False	(unchanged)	–	Toggle for redshift corrections (reserved, unused)
<i>Cosmological model</i>				

Symbol	Default	First value	Unit	Meaning
$\mathcal{R}$	260	(unchanged)	$\text{Gpc}^{-3} \text{yr}^{-1}$	Volumetric GRB rate density
D_{euc}	$1.404 \times 10^{28} \text{ cm} \approx 4.55 \text{ Gpc}$	$1.63 \times 10^{28} \text{ cm} \approx 5.28 \text{ Gpc}^*$	cm	Euclidean calibration distance
R_{int}	$\frac{4\pi}{3} \mathcal{R} D_{\text{euc}}^3 \approx 1.03 \times 10^5$	$\approx 1.60 \times 10^5^*$	yr^{-1}	Total GRBs/yr in calibration volume
<i>Survey instrument</i> (<code>SurveyTelescopeParams</code> , <code>SurveyDesignParams</code>)				
F_{ref}	$10^{-4.68} \approx 2.09 \times 10^{-5}$	(unchanged)	Jy	Limiting flux anchor at $t_{\text{exp,ref}}$
α	0.5	(unchanged)	–	Flux–exposure exponent (background-limited)
Ω_{exp}	47	(unchanged)	deg^2	Field of view per exposure
$t_{\text{exp,ref}}$	30	(unchanged)	s	Reference exposure time for the F_{ref} anchor
$\Omega_{\text{srv,max}}$	27500 [†]	(unchanged)	deg^2	Maximum surveyable sky area
t_{night}	36000 s (= 10 hr)	(unchanged)	s	Astronomical night duration
<i>Survey strategy</i> (free variables)				
N_{exp}	–	–	–	Exposures per cadence cycle
t_{cad}	–	–	s	Cadence interval
f_{live}	0.1 [†]	(unchanged)	–	Fraction of t_{cad} spent observing
t_{OH}	0	(unchanged)	s	Fixed overhead per exposure
i	2 [‡]	10 ^{*‡}	–	Required detections per GRB
<i>Constraints</i> (detection-rate filters)				
q_{min}	0	0 (introduced 2026-05-15)	–	Minimum viewing-angle parameter; 0 = no filter
D_{min}	0	0 (introduced 2026-05-15)	Gpc	Minimum detection distance; 0 = no filter
s_{fade}	0	0 (introduced 2026-07, as <code>s_min</code>)	mag/day	Minimum fading-rate identification cut; 0 = no filter
s_{rise}	0	0 (introduced 2026-07)	mag/day	Minimum rise-rate identification cut; 0 = no filter
<i>Luminosity function</i> (Section 9; off by default)				
α	–2	–2	–	Index of $d\mathcal{R}/dL \propto L^\alpha$
$\mathcal{A}$	1	1	$\text{Gpc}^{-3} \text{yr}^{-1}$	Rate density per $d \ln L$ at L_{ref}
L_{ref}	10^{32}	fixed	$\text{erg s}^{-1} \text{Hz}^{-1}$	Reference for $L_\nu(t_{\text{dec}})$

* **Changed since introduction** (`grb_detect/params.py`'s history, first committed as 684f9c5, 2026-02-20): p 2.5→2.2 (this update); ε_B 0.01 → $10^{-3.4}$ (76eaa31, 2026-07-20) → 10^{-4} (a later update); D_{euc} 5.28→4.55 Gpc, carrying R_{int} with it (76eaa31, 2026-07-20); i 10→2 (76eaa31, 2026-07-20). All other rows are unchanged from the first commit.

[†] The interactive UI intentionally ships different initial slider values than these library defaults: $\Omega_{\text{srv,max}} = 41253 \text{ deg}^2$ (full sky) and $f_{\text{live}} = 0.2$. The library defaults above are what a bare constructor call (`SurveyDesignParams()`, `SurveyTelescopeParams()`) still uses.

[‡] i has no library-level default anywhere in `grb_detect/params.py` (always a required argument); the values shown are the interactive UI's initial slider position instead — first the original Dash app's `dcc.Slider` default, now `build_standalone.py`'s slider default.

Note on R_{int} : The `AfterglowPhysicalParams` dataclass bakes R_{int} in at class-load time from the default $\mathcal{R}$ and D_{euc} values. When constructing the class with non-default $\mathcal{R}$ or D_{euc} , always recompute R_{int} explicitly and pass it to the constructor:

$$R_{\text{int}} = \frac{4\pi}{3} \mathcal{R} D_{\text{euc,Gpc}}^3$$

2 Afterglow Model (ISM)

2.1 Deceleration and jet-break times

Blandford–McKee ISM dynamics (thesis Eq. 30):

$$t_{\text{dec}} = 27 \zeta E_{52}^{1/3} n_0^{-1/3} \Gamma_{2.5}^{-8/3} \text{ [s]}, \quad \zeta = \frac{1}{3} \text{ (Euclidean approximation, } z \rightarrow 0) \quad (1)$$

where $E_{52} = E_{k,\text{iso}}/10^{52} \text{ erg}$ and $\Gamma_{2.5} = \Gamma_0/10^{2.5}$.

$$t_j = t_{\text{dec}} (\Gamma_0 \theta_j)^{8/3} \quad (2)$$

2.2 Viewing-angle parameterisation

Define normalized viewing angle $q = \theta_{\text{obs}}/\theta_j$ and $\tilde{q} = q - 1$. Three characteristic boundaries separate the afterglow into phases:

$$q_{\text{dec}} = 1 + (\Gamma_0 \theta_j)^{-1}, \quad q_j = 2, \quad q_{\text{nr}} = \frac{\sqrt{2}}{\theta_j} \quad (3)$$

Phase I ($t < t_{\text{dec}}$) is excluded by design. Phase II: $q_{\text{dec}} \leq q < q_j$. Phase III: $q \geq q_j$.

Beaming identity. The normalization $q_{\text{nr}} = \sqrt{2}/\theta_j$ is chosen so that

$$\frac{\theta_j^2}{2} q_{\text{nr}}^2 = 1,$$

i.e. the integral over all observable solid angles $\int_0^{q_{\text{nr}}} \theta_j^2 q dq = 1$ exactly. This ensures that in regime A1 (saturation), $R_{\text{A1}} = f_{\Omega} R_{\text{int}}$ without any additional beaming prefactor. For all other regimes the factor $\frac{1}{2} \theta_j^2 q_{\text{max}}^2 < 1$ represents the fraction of GRBs detectable at viewing angles $q \leq q_{\text{max}}$.

2.3 Off-axis peak flux (PLS G, $\nu_m < \nu < \nu_c$)

On-axis reference flux at t_{dec} (thesis Eq. 24):

$$F_{\text{dec}}^{\text{G}} = \frac{0.461}{1000} (p - 0.04) e^{2.53p} \bar{\varepsilon}_e^{p-1} \varepsilon_B^{(p+1)/4} n_0^{1/2} E_{52}^{(p+3)/4} t_{\text{dec,d}}^{-3(p-1)/4} D_{28}^{-2} \nu_{14}^{-(p-1)/2} \text{ [Jy]} \quad (4)$$

where $\bar{\varepsilon}_e = \varepsilon_e(p-2)/(p-1)$, $t_{\text{dec,d}} = t_{\text{dec}}/\text{day}$, $D_{28} = D_{\text{euc}}/10^{28} \text{ cm}$, $\nu_{14} = \nu/10^{14} \text{ Hz}$. Valid for $p > 2$.

Off-axis flux exponents (PLS G):

$$a_{\text{II}}^{\text{G}} = p - 1, \quad a_{\text{III}}^{\text{G}} = p.$$

These follow from the afterglow temporal decay index $F \propto t^{-3(p-1)/4}$ in Phase II (combined with $t_p \propto \tilde{q}^{8/3}$), and $F \propto t^{-p}$ after the jet break (Phase III, with $t_p \propto \tilde{q}^2$).

2.4 Off-axis reference flux (PLS H, $\nu > \nu_c$)

For PLS H (thesis Eq. 25), the on-axis reference flux is:

$$F_{\text{dec}}^{\text{H}} = \frac{0.855}{1000} (p - 0.98) e^{1.95p} \bar{\varepsilon}_e^{p-1} \varepsilon_B^{(p-2)/4} E_{52}^{(p+2)/4} t_{\text{dec,d}}^{-(3p-2)/4} D_{28}^{-2} \nu_{14}^{-p/2} \text{ [Jy]} \quad (5)$$

Key differences from PLS G: no n_0 dependence; ε_B exponent $(p-2)/4$; E_{52} exponent $(p+2)/4$; time exponent $-(3p-2)/4$; ν exponent $-p/2$.

Off-axis flux exponents (PLS H):

$$a_{\text{II}}^{\text{H}} = \frac{3p-2}{3}, \quad a_{\text{III}}^{\text{H}} = p.$$

The Phase II exponent follows from temporal slope $-(3p-2)/4$ combined with $t_p \propto \tilde{q}^{8/3}$ in Phase II.

2.5 Off-axis peak flux scaling

The off-axis peak flux at viewing angle q relative to the on-axis value F_{dec} is:

$$\frac{F_p(q)}{F_{\text{dec}}} = \begin{cases} \left(\frac{\tilde{q}}{\tilde{q}_{\text{dec}}}\right)^{-2a_{\text{II}}} & q_{\text{dec}} \leq q < q_j \text{ (Phase II)} \\ \tilde{q}_{\text{dec}}^{-2} \left(\frac{\tilde{q}}{\tilde{q}_{\text{dec}}}\right)^{-2a_{\text{III}}} & q \geq q_j \text{ (Phase III)} \end{cases} \quad (6)$$

Reference fluxes at the characteristic boundaries (used to select the q_{Euc} branch):

$$F_j = F_{\text{dec}} \tilde{q}_{\text{dec}}^{2a_{\text{II}}}, \quad F_{\text{nr}} = F_{\text{dec}} \tilde{q}_{\text{dec}}^{-2} \left(\frac{\tilde{q}_{\text{nr}}}{\tilde{q}_{\text{dec}}}\right)^{-2a_{\text{III}}}.$$

At the boundary $q = q_j$ ($\tilde{q} = 1$), the Phase II and Phase III expressions are continuous for PLS G (since $a_{\text{III}}^{\text{G}} - 1 = a_{\text{II}}^{\text{G}}$) but not for PLS H.

3 Survey Model

$$t_{\text{exp}} = \frac{f_{\text{live,eff}} t_{\text{cad}}}{N_{\text{exp}}} - t_{\text{OH}}, \quad F_{\text{lim}} = F_{\text{ref}} \left(\frac{t_{\text{exp}}}{t_{\text{ref}}}\right)^{-\alpha}, \quad f_{\Omega} = \frac{N_{\text{exp}} \Omega_{\text{exp}}}{4\pi} \quad (7)$$

The effective live fraction $f_{\text{live,eff}}$ depends on the cadence regime:

$$f_{\text{live,eff}} = \begin{cases} f_{\text{live}} & t_{\text{cad}} = n \cdot 86400 \text{ s}, n \in \mathbb{N} \quad (\text{discrete, per-day}) \\ f_{\text{live}}/f_{\text{night}} & t_{\text{cad}} < 86400 \text{ s and optical-survey mode} \quad (\text{continuous, sub-night}) \end{cases}$$

with $f_{\text{night}} = t_{\text{night}}/86400$. In the continuous regime all exposures land inside the night window of length t_{night} , so the within-night live fraction is $f_{\text{live}} \cdot 86400/t_{\text{night}} = f_{\text{live}}/f_{\text{night}}$. The total detection rate on the continuous branch is additionally multiplied by f_{night} to account for the nighttime accessibility of GRBs. The construction is physical only when $f_{\text{night}} \geq f_{\text{live}}$ (so $f_{\text{live,eff}} \leq 1$); this is enforced in the UI by a dynamic lower bound on the t_{night} slider.

A strategy point $(N_{\text{exp}}, t_{\text{cad}})$ is in the valid region A0 when $t_{\text{exp}} > 0$, $N_{\text{exp}} \geq 1$, and $N_{\text{exp}} \leq N_{\text{exp,max}} \equiv \Omega_{\text{srv,max}}/\Omega_{\text{exp}}$.

4 Detection Rate: Two Horizons and Seven Regimes

4.1 Distance and horizon angles

The maximum detection distance for an on-axis burst ($q=0$):

$$D_{\text{dec}} = D_{\text{euc}} \left(\frac{F_{\text{lim}}}{F_{\text{dec}}}\right)^{-1/2} \quad (8)$$

Euclidean horizon angle q_{Euc} : the largest off-axis angle detectable at the calibration distance D_{euc} :

$$\tilde{q}_{\text{Euc}} = \begin{cases} \tilde{q}_{\text{dec}} \left(\frac{F_{\text{lim}}}{F_{\text{dec}}}\right)^{-1/(2a_{\text{II}})} & F_{\text{lim}} \geq F_j \text{ (Phase II)} \\ \tilde{q}_{\text{dec}}^{(a_{\text{III}}-1)/a_{\text{III}}} \left(\frac{F_{\text{lim}}}{F_{\text{dec}}}\right)^{-1/(2a_{\text{III}})} & F_{\text{lim}} < F_j \text{ (Phase III)} \end{cases} \quad (9)$$

The two branches are continuous at $F_{\text{lim}} = F_j$ (for PLS G), where both give $\tilde{q}_{\text{Euc}} = 1$ (i.e. $q_{\text{Euc}} = q_j$).

Off-axis peak time. A burst at viewing angle q reaches its afterglow flux peak at time:

$$t_p(q) = \begin{cases} t_j \tilde{q}^{8/3} & q_{\text{dec}} \leq q < q_j \text{ (Phase II)} \\ t_j \tilde{q}^2 & q \geq q_j \text{ (Phase III)} \end{cases} \quad (10)$$

Both expressions equal t_j continuously at $\tilde{q} = 1$ ($q = q_j$). At $q = q_{\text{dec}}$ ($\tilde{q} = \tilde{q}_{\text{dec}}$), Phase II gives $t_p = t_{\text{dec}}$ as expected.

Cadence horizon angle q_i : largest off-axis angle whose peak arrives within i cadence intervals ($t_p = i t_{\text{cad}}$):

$$\tilde{q}_i = \begin{cases} (i t_{\text{cad}}/t_j)^{3/8} & i t_{\text{cad}} < t_j \text{ (Phase II peak)} \\ (i t_{\text{cad}}/t_j)^{1/2} & i t_{\text{cad}} \geq t_j \text{ (Phase III peak)} \end{cases} \quad (11)$$

Cadence distance D_i : maximum detection distance at angle q_i :

$$D_i = \begin{cases} D_{\text{dec}} (\tilde{q}_i/\tilde{q}_{\text{dec}})^{-a_{\text{II}}} & i t_{\text{cad}} < t_j \\ (D_{\text{dec}}/\tilde{q}_{\text{dec}}) (\tilde{q}_i/\tilde{q}_{\text{dec}})^{-a_{\text{III}}} & i t_{\text{cad}} \geq t_j \end{cases} \quad (12)$$

4.2 Piecewise rate formulas (dominant-term approximation)

Regime	Condition	Rate R_k [yr^{-1}]
A1	$q_{\text{Euc}} \geq q_{\text{nr}}$ and $q_{\text{Euc}} \geq q_i$	$f_{\Omega} R_{\text{int}}$
A2	$q_j \leq q_{\text{Euc}} < q_{\text{nr}}$ and $q_{\text{Euc}} \geq q_i$	$\frac{1}{2} f_{\Omega} \theta_j^2 q_{\text{Euc}}^2 R_{\text{int}}$
A3	$q_{\text{dec}} \leq q_{\text{Euc}} < q_j$ and $q_{\text{Euc}} \geq q_i$	$\frac{1}{2} f_{\Omega} \theta_j^2 q_{\text{Euc}}^2 R_{\text{int}}$
A4	$q_i \geq q_{\text{nr}}$ and $q_{\text{Euc}} < q_i$	$\frac{1}{2} f_{\Omega} \theta_j^2 q_{\text{nr}}^2 (D_i/D_{\text{euc}})^3 R_{\text{int}}$
A5	$q_j \leq q_i < q_{\text{nr}}$ and $q_{\text{Euc}} < q_i$	$\frac{1}{2} f_{\Omega} \theta_j^2 q_i^2 (D_i/D_{\text{euc}})^3 R_{\text{int}}$
A6	$q_{\text{dec}} \leq q_i < q_j$ and $q_{\text{Euc}} < q_i$	$\frac{1}{2} f_{\Omega} \theta_j^2 q_i^2 (D_i/D_{\text{euc}})^3 R_{\text{int}}$
A7	$q_{\text{Euc}} < q_{\text{dec}}$ and $q_i < q_{\text{dec}}$	$\frac{1}{2} f_{\Omega} \theta_j^2 q_{\text{dec}}^2 (D_{\text{dec}}/D_{\text{euc}})^3 R_{\text{int}}$

Regimes A2 and A3 share the same formula (both flux-limited); the distinction is only which afterglow phase q_{Euc} falls in. Similarly A5 and A6 share the same formula (both cadence-limited). The general structure is:

$$R_k = \frac{1}{2} f_{\Omega} \theta_j^2 q_{\text{max}}^2 (D_{\text{eff}}/D_{\text{euc}})^3 R_{\text{int}},$$

where q_{max} is the relevant upper angular limit and D_{eff} is the corresponding detection distance. The output surface shows $\log_{10} R_{\text{det}}$.

Region A0 ($t_{\text{exp}} \leq 0$, $N_{\text{exp}} < 1$, or $N_{\text{exp}} > N_{\text{exp,max}}$) is undefined (NaN). Boundary comparisons use tolerance $\varepsilon = 10^{-12}$.

5 Full-Integral Mode (Optional)

Instead of the dominant-term formulas, the rate is computed by numerically integrating over all viewing angles using normalized (dimensionless) distances $\tilde{D} = D/D_{\text{euc}}$:

$$R = f_{\Omega} \theta_j^2 R_{\text{int}} \int_0^{q_{\text{nr}}} q [\min(\tilde{D}_{\text{max}}(q), \tilde{D}_{\text{eff}})]^3 dq, \quad (13)$$

where $\tilde{D}_{\text{eff}} = \min(D_i/D_{\text{euc}}, 1)$ encodes the cadence constraint and the Euclidean horizon (under the `win_from_peak` setting of Sec. 7 it becomes the q -dependent $\min(\tilde{D}_w(q), 1)$ of Eq. (29)), and

$$\tilde{D}_{\text{max}}(q) = \begin{cases} \tilde{D}_{\text{dec}} & q < q_{\text{dec}} \\ \tilde{D}_{\text{dec}}(\tilde{q}/\tilde{q}_{\text{dec}})^{-a_{\text{II}}} & q_{\text{dec}} \leq q < q_j \\ (\tilde{D}_{\text{dec}}/\tilde{q}_{\text{dec}})(\tilde{q}/\tilde{q}_{\text{dec}})^{-a_{\text{III}}} & q \geq q_j \end{cases} \quad (14)$$

with $\tilde{D}_{\text{dec}} = (F_{\text{lim}}/F_{\text{dec}})^{-1/2}$ (dimensionless). Both integrands are dimensionless, so (13) has units of R_{int} [yr^{-1}] as required.

The dominant-term approximation (Sec. 4.2) is the leading term of (13) evaluated at the boundary angle (q_{Euc} or q_i). The full integral adds contributions from all other angles. Evaluated with a trapezoidal rule over $N_q = 500$ points ($< 0.3\%$ error for typical parameters).

6 Detection-Rate Filters: q_{min} and D_{min}

The model exposes two physics filters that restrict which GRBs contribute to the detection rate. They replace an earlier “off-axis only” boolean toggle (deprecated; see Sec. 6.7).

Definitions.

- $q_{\text{min}} \in [0, q_{\text{nr}}]$: minimum viewing-angle parameter. Only GRBs with $q \geq q_{\text{min}}$ count toward the rate. Default $q_{\text{min}} = 0$ (no filter).
- $D_{\text{min}} \in [0, D_{\text{euc}}]$: minimum *detectable* distance. The integration shell is restricted to $D \geq D_{\text{min}}$. Default $D_{\text{min}} = 0$ (no filter).

Interpretation note. D_{min} is a *detectability-distance* floor, not a source-distance floor. The detector volume integral is performed over the shell $[D_{\text{min}}, D_{\text{eff}}(q)]$, where $D_{\text{eff}}(q)$ is the maximum distance at which a burst at angle q is detectable. Bursts whose maximum detectable distance $D_{\text{eff}}(q)$ falls below D_{min} are excluded.

6.1 Joint distribution of detected bursts

For uniform GRB number density and isotropic jet axes, the joint detection PDF on the support $\{q \in [0, q_{\text{nr}}], 0 \leq D \leq D_{\text{eff}}(q)\}$ is

$$p(q, D) \propto q \cdot D^2,$$

combining the solid-angle factor $\theta_j^2 q dq$ and the Euclidean volume element $D^2 dD$. The unfiltered total rate is

$$R = f_{\Omega} \theta_j^2 R_{\text{int}} \int_0^{q_{\text{nr}}} q \frac{1}{3} D_{\text{eff}}(q)^3 dq,$$

in agreement with Eq. (13) after absorbing the constant $\frac{1}{3}$ into the rate normalization.

6.2 Filtered dominant-term formula

When $D_{\text{eff}}(q)$ is constant over the integration range $[q_{\text{min}}, q_{\text{max}, k}]$ (which holds by construction in each regime k), the inner D -integral becomes

$$\int_{D_{\text{min}}}^{D_{\text{eff}, k}} D^2 dD = \frac{1}{3} [D_{\text{eff}, k}^3 - D_{\text{min}}^3]_+,$$

where $[x]_+ \equiv \max(x, 0)$ enforces non-negativity when the filter excludes the entire shell. The q -integral yields the same algebraic form:

$$\int_{q_{\text{min}}}^{q_{\text{max}, k}} q dq = \frac{1}{2} [q_{\text{max}, k}^2 - q_{\text{min}}^2]_+.$$

Combining, the filtered regime rate is

$$R_k(q_{\min}, D_{\min}) = \frac{1}{2} f_{\Omega} \theta_j^2 R_{\text{int}} [q_{\max, k}^2 - q_{\min}^2]_+ [(D_{\text{eff}, k}/D_{\text{euc}})^3 - (D_{\min}/D_{\text{euc}})^3]_+. \quad (15)$$

At $q_{\min} = D_{\min} = 0$ this reduces exactly (to floating-point) to the unfiltered formula in Sec. 4.2. When either bracket evaluates to zero, the regime contributes nothing and the cell's $\log_{10} R$ is NaN.

6.3 Filtered full-integral formula

In full-integral mode the inner D -integral cannot be evaluated in closed form because $D_{\text{eff}}(q)$ is piecewise:

$$R = f_{\Omega} \theta_j^2 R_{\text{int}} \int_0^{q_{\text{nr}}} \mathbf{1}\{q \geq q_{\min}\} q [D_{\text{eff}}(q)^3 - (D_{\min}/D_{\text{euc}})^3]_+ dq, \quad (16)$$

where $D_{\text{eff}}(q) = \min(\tilde{D}_{\max}(q), \tilde{D}_{\text{eff}})$ is the same profile defined in Eq. (14). The indicator $\mathbf{1}\{q \geq q_{\min}\}$ is applied to a discrete q -grid of $N_q = 500$ points; the trapezoidal rule near $q = q_{\min}$ introduces an $\mathcal{O}(\Delta q/q_{\text{nr}}) \approx 2 \times 10^{-3}$ relative bias. This bias is verified empirically by the test `test_B12_indicator_bias_at_qmin_bounded`.

6.4 Filtered medians

Under the dominant-term assumption (constant D_{eff} on the regime), the marginals are

$$\begin{aligned} p(q) &\propto q \cdot [D_{\text{eff}}^3 - D_{\min}^3]_+ \propto q, \quad q \in [q_{\min}, q_{\max}], \\ p(D) &\propto D^2 \cdot [q_{\max}^2 - q_{\min}^2]_+ \propto D^2, \quad D \in [D_{\min}, D_{\text{eff}}]. \end{aligned}$$

Solving $F(q_{\text{med}}) = \frac{1}{2}$ for each gives

$$q_{\text{med}} = \sqrt{\frac{q_{\max}^2 + q_{\min}^2}{2}}, \quad D_{\text{med}} = \left(\frac{D_{\text{eff}}^3 + D_{\min}^3}{2} \right)^{1/3}. \quad (17)$$

At $q_{\min} = D_{\min} = 0$ these reduce to $q_{\max}/\sqrt{2}$ and $D_{\text{eff}} \cdot 2^{-1/3}$ (Sec. 6.1). When the filter excludes all weight ($q_{\max} \leq q_{\min}$ or $D_{\text{eff}} \leq D_{\min}$), both medians return NaN.

In full-integral mode where $D_{\text{eff}}(q)$ varies, the q -marginal becomes

$$p(q) \propto q \cdot [D_{\text{eff}}(q)^3 - (D_{\min}/D_{\text{euc}})^3]_+ \cdot \mathbf{1}\{q \geq q_{\min}\},$$

which is *not* simply $\propto q$, so the analytic formula (17) no longer applies; the median is found by cumulative trapezoidal integration of this weighted form. See `compute_medians_numerical`.

6.5 Fading-rate filter s_{fade}

Real optical transient surveys filter their alert streams to “afterglow-like” candidates by requiring a measured magnitude decline faster than some threshold s_{fade} in mag/day; common values are 0.3 (ZTF-style) and 1 (stricter). The filter is exposed as a third sidebar parameter alongside $q_{\min}$ and $D_{\min}$, with two modes that coincide in the $t_{\text{cad}} \rightarrow 0$ limit.

Setup. Post-peak, the lightcurve obeys $F(t) \propto t^{\alpha}$ with $\alpha < 0$, so

$$m(t) = -2.5\alpha \log_{10}(t/t_p) + \text{const}, \quad \frac{dm}{dt} = -\frac{2.5\alpha}{t \ln 10} \text{ mag/s}.$$

The survey samples this lightcurve starting at the first post-peak detection $t_{\text{first}} \geq t_p$ and ending at $t_{\text{last}} = t_{\text{first}} + (i-1)t_{\text{cad}}$. A key identity: for a pure power law the measured slope between t_{first} and t_{last} depends only on t_{first} and the baseline $(i-1)t_{\text{cad}}$ — not on t_p separately — so the cut inverts cleanly to a cap on the *first-detection time*.

Discrete (per-cadence) mode. The measured slope across the i detections is

$$s_{\text{meas}} = -2.5 \alpha \log_{10} \left(1 + \frac{(i-1)t_{\text{cad}}}{t_{\text{first}}} \right) \frac{\text{DAY_S}}{(i-1)t_{\text{cad}}} \text{ [mag/day]}. \quad (18)$$

Imposing $s_{\text{meas}} \geq s_{\text{fade}}$ and solving for the latest t_{first} that still passes, with $A = 2.5|\alpha|$ and $E = s_{\text{fade}}(i-1)t_{\text{cad}}/(\text{DAY_S} \cdot A)$:

$$t_{f,s} = \frac{(i-1)t_{\text{cad}}}{10^E - 1}, \quad \text{pass} \iff t_{\text{first}} \leq t_{f,s}. \quad (19)$$

This is strategy-dependent: long t_{cad} shrinks $t_{f,s}$ (the survey can no longer resolve the slope). At $i = 1$ the slope is undefined and the cut is bypassed.

Continuous (instantaneous) mode. Evaluate $|dm/dt|$ at the first detection directly and require it to exceed s_{fade} :

$$t_{f,s} = \frac{2.5|\alpha| \text{DAY_S}}{s_{\text{fade}} \ln 10}. \quad (20)$$

Strategy-independent; defined for all i .

Limit equivalence. As $t_{\text{cad}} \rightarrow 0$, $E \rightarrow 0$ and $10^E - 1 \approx E \ln 10$, so (19) reduces to (20):

$$t_{f,s} \rightarrow \frac{(i-1)t_{\text{cad}}}{E \ln 10} = \frac{2.5|\alpha| \text{DAY_S}}{s_{\text{fade}} \ln 10}.$$

The continuous mode is the high-cadence limit of the discrete mode.

Dominant-term treatment: best-case start. In the closed-form (dominant-term) rate mode we assume the first detection lands at the peak, $t_{\text{first}} \approx t_p(\tilde{q})$, so $t_{f,s}$ caps the peak time itself. The model's q_i relation (§ Viewing-angle parameterisation) gives $t_p(\tilde{q}) = t_j \tilde{q}^{8/3}$ in Phase II and $t_j \tilde{q}^2$ in Phase III. Inverting:

$$\tilde{q}_{s,\text{II}} = \left(\frac{t_{f,s,\text{II}}}{t_j} \right)^{3/8}, \quad \tilde{q}_{s,\text{III}} = \left(\frac{t_{f,s,\text{III}}}{t_j} \right)^{1/2}, \quad (21)$$

where α in $t_{f,s}$ is the phase-specific temporal slope: $\alpha_{\text{II}} = -3(p-1)/4$, $\alpha_{\text{III}} = -p$ for PLS G, and $\alpha_{\text{II}} = -(3p-2)/4$, $\alpha_{\text{III}} = -p$ for PLS H. The cut then acts on each regime's effective q_{max} as a phase-mapped hard cap: the seven dominant-term rates (15) become

$$r_k = \frac{1}{2} f_{\Omega} \theta_j^2 R_{\text{int}} \cdot [\min(q_{\text{max},k}, q_{s,\Phi(k)})^2 - q_{\text{min}}^2]_+ \cdot [(D_{\text{eff},k}/D_{\text{euc}})^3 - (D_{\text{min}}/D_{\text{euc}})^3]_+,$$

with phase assignment Φ : A1, A2, A4, A5 $\rightarrow$ Phase III; A3, A6, A7 $\rightarrow$ Phase II. As shown next, this best-case treatment is the exact $t_{\text{cad}} \rightarrow 0$ limit of the uniform-start model and an optimistic upper bound otherwise — consistent with the dominant-term philosophy of keeping the rate expressions closed-form.

Uniform observation start (full-integral mode). The survey's visit schedule has a random phase relative to the burst, so the first post-peak visit actually lands at

$$t_{\text{first}} = t_p(\tilde{q}) + u, \quad u \sim \mathcal{U}(0, t_{\text{cad}}).$$

Since $\text{pass} \iff t_{\text{first}} \leq t_{f,s}$, each burst passes the cut with probability

$$P(q) = \text{clip} \left(\frac{t_{f,s} - t_{p,\text{eff}}(q)}{t_{\text{cad}}}, 0, 1 \right), \quad t_{p,\text{eff}}(q) = \max(t_j [\max(q-1, 0)]^\beta, t_{\text{dec}}), \quad (22)$$

with $\beta = 8/3$ (Phase II) and $\beta = 2$ (Phase III). P reaches 0 exactly at the hard cap q_s of (21) and ramps up (linearly in t_p) below it. The t_{dec} floor on the effective peak time is *exact* for on-axis viewers — $t_j \tilde{q}_{\text{dec}}^{8/3} = t_{\text{dec}}$ identically, so the floor binds precisely for $q \leq q_{\text{dec}}$ (where the model lightcurve peaks at t_{dec}) and is a no-op above; on-axis viewers are lumped with Phase II as in the existing masks. For the fade weight the floor matters only at $\mathcal{O}(t_{\text{dec}}/t_{\text{cad}}) \approx 19 \text{ s}/t_{\text{cad}}$ (it is however *essential* for the rise filter, Sec. 6.6); one physical consequence: when $t_{f,s} < t_{\text{dec}}$ — i.e. even a burst caught at the earliest possible epoch fades too slowly — the weight now correctly annihilates the rate instead of leaving the fictitious pre- t_{dec} residual of the pure extrapolation. In the full-integral form (16) the integrand simply picks up the factor $P(q)$ in place of the former indicator $\mathbf{1}\{q \leq q_{s,\Phi(q)}\}$; this weight is applied only where the integral is evaluated numerically anyway (`rate_log10_full_integral`, the numerical medians, and the dR/dq , dR/dD views), leaving the closed-form rate equations untouched.

Note the qualitatively new effect: when $t_{f,s} < t_{\text{cad}}$ even on-axis bursts pass only with probability $t_{f,s}/t_{\text{cad}}$ — with a long cadence the afterglow is usually caught late, when its mag/day decline is slow. The best-case hard cap would let all of them pass.

Limits and invariants. (i) $s_{\text{fade}} \rightarrow 0$: $t_{f,s} = +\infty$, $P \equiv 1$, and both rate modes collapse to the unfiltered baseline to floating point. (ii) $t_{\text{cad}} \rightarrow 0$: the ramp width vanishes and $P \rightarrow \mathbf{1}\{q \leq q_s\}$, recovering the dominant-term hard cap; the deficit is linear in t_{cad} . (iii) $P(q) \leq \mathbf{1}\{q \leq q_s\}$ pointwise, so under an active cut the full-integral rate sits *below* the dominant-term rate — an expected, physical mode disagreement, not a numerical artefact. (iv) $t_{f,s}$ is strictly decreasing in s_{fade} , so both treatments remain monotone non-increasing in s_{fade} .

Assumptions. (a) The start phase u is treated as independent of the detection condition itself; the q_i detection machinery keeps its dominant-term form. This is coherent: $u \leq t_{\text{cad}}$ implies $t_{\text{last}} \leq t_p + i t_{\text{cad}}$, matching the q_i detection window. (b) The rise before the peak is ignored — the slope is measured from the first post-peak detection. (c) The measurement segment is treated as lying in a single power-law phase; a delayed segment that straddles t_j actually fades *faster* than modeled (the post-break slope is steeper), so the treatment is conservative.

Implementation reference. `DetectionRateModel._t_first_caps()` returns the phase-resolved $t_{f,s,\text{II}}$ and $t_{f,s,\text{III}}$ arrays (broadcastable to the strategy grid). `_q_s_fading_caps()` maps them to the best-case hard caps (21) used by `rate_log10` and `compute_medians_analytic`; `_fading_survival()` evaluates the pointwise $P(q)$ of (22) used by `rate_log10_full_integral`, `compute_medians_numerical`, `dR_dq_full_integral`, and `dR_dD_full_integral`. Since P is a continuous weight rather than a cut, the numerical median integrals remain valid without any non-contiguous-interval special-casing.

6.6 Rise-rate filter s_{rise}

Alert streams also cut on the *rise*: a new candidate is credible when the previous visit — one cadence before the first detection — was a non-detection, bounding the earlier magnitude by the limiting magnitude m_{lim} . The implied rise rate is then at least $(m_{\text{lim}} - m_{\text{first}})/t_{\text{cad}}$, and the filter requires it to exceed s_{rise} [mag/day].

Semantics. With $m = -2.5 \log_{10} F + \text{const}$,

$$\frac{m_{\text{lim}} - m_{\text{first}}}{t_{\text{cad}}/\text{DAY}_S} \geq s_{\text{rise}} \iff F(t_{\text{first}}) \geq \eta F_{\text{lim}}, \quad \eta = 10^E, \quad E = \frac{s_{\text{rise}} t_{\text{cad}}}{2.5 \text{ DAY}_S}. \quad (23)$$

The boost η grows *exponentially* with cadence: a slow survey can only certify a fast riser if the source jumped far above the limit since the last visit. The model ignores the pre-peak rise, so

the previous visit (always pre-peak, $t_{\text{first}} - t_{\text{cad}} < t_p$) is automatically a non-detection — the filter is internally consistent with the baseline. Unlike the discrete fade filter, the rise cut applies at $i_{\text{det}} = 1$ (it needs only the previous visit and the first detection); it has no discrete/continuous mode split — the measurement *is* a previous-visit finite difference, so `s_mode` remains fade-only.

Dominant-term treatment: best-case start. With the first detection at the peak, pass $\iff F_p(q, D) \geq \eta F_{\text{lim}} \iff D \leq D_{\text{max}}(q)/\sqrt{\eta}$. Because the flux-limited boundary functions take F_{lim} as an argument, the raised-limit boundaries are direct evaluations:

$$q_{E,r} = q_{\text{Euc}}(\eta F_{\text{lim}}), \quad q_{r,i} = q_{\text{Euc}}(\eta F_{\text{lim}} (D_i/D_{\text{euc}})^2), \quad D_{\text{dec},r} = D_{\text{dec}} \eta^{-1/2},$$

where $q_{r,i}$ is the angle at which $D_{\text{max}}(q)/\sqrt{\eta} = D_i$; since $D_i = D_{\text{max}}(q_i)$ identically, $q_{r,i}(\eta=1) = q_i$ exactly. The caps enter the regime formulas as $q_{\text{max}} \rightarrow \min(q_{\text{max}}, q_{E,r})$ for the flux-limited A1–A3, $\min(q_{\text{max}}, q_{r,i})$ for the cadence-limited A4–A6, and $D_{\text{eff},7} \rightarrow \min(D_{\text{dec}}, D_i, D_{\text{dec},r})$ for A7, composing with the fading caps of (21) by further $\min()$.

Deep-cut correctness (on-axis max-term). q_{Euc} extrapolates the off-axis power law and never returns $q < 1$, so for a deep cut the truncated angular term would saturate at $[1 - q_{\text{min}}^2]_+ \times D$ -factor instead of vanishing. Moreover the extrapolated boundary is the *true* flux boundary only while it sits above q_{dec} : below the crossing, the on-axis $D_{\text{max}} = D_{\text{dec}}$ plateau makes the angular box overcount. Each of r_1 – r_6 is therefore computed as the **max of two closed-form lower bounds** of the true pass volume:

$$r_k = \text{base} \cdot \max \left\{ \underbrace{\mathbf{1}\{q_{\text{cap},k} \geq q_{\text{dec}}\} [q_{\text{max},k,\text{eff}}^2 - q_{\text{min}}^2]_+ [\tilde{D}_{\text{eff},k}^3 - \tilde{D}_{\text{min}}^3]_+}_{\text{angular term (valid above the crossing)}}, \underbrace{[\min(q_{\text{dec}}, q_{\text{max},k}, q_{s,\text{II}})^2 - q_{\text{min}}^2]_+ [\min(\tilde{D}_{\text{eff},k}, \tilde{D}_{\text{dec}} \eta^{-1/2})^3 - \tilde{D}_{\text{min}}^3]_+}_{\text{on-axis term}} \right\}.$$

At the crossing $q_{\text{cap},k} = q_{\text{dec}}$ one has $D_{\text{dec}}/\sqrt{\eta} = D_{\text{eff-scale}}$, so the two terms are equal and the max is continuous in s_{rise} ; at $\eta = 1$ the on-axis term is dominated and the baseline is recovered; in the deep-cut limit $r_k \propto \eta^{-3/2} \rightarrow 0$ (verified numerically over 6 dex).

Uniform observation start (full-integral mode). Post-peak, $F(t) = F_p(t/t_p)^{-|\alpha|}$ crosses the raised limit at

$$t_{\text{lim}}(q, \tilde{D}) = t_{p,\text{eff}}(q) \left(\frac{\tilde{D}_0(q)}{\tilde{D}} \right)^k, \quad k = \frac{2}{|\alpha_\Phi|}, \quad \tilde{D}_0(q) = \tilde{D}_{\text{max}}(q) \eta^{-1/2}, \quad (24)$$

so pass $\iff t_{\text{first}} \leq t_{\text{lim}}$. Here the t_{dec} floor of $t_{p,\text{eff}}$ (22) is *essential*: $t_{\text{lim}} \propto t_{p,\text{eff}}$, and the pure extrapolation ($t_p \rightarrow 0$ on-axis) would wrongly annihilate all on-axis rise weights; with the floor, the general formula reproduces the on-axis lightcurve $F = F_{\text{dec}}(D) (t/t_{\text{dec}})^{-|\alpha_{\text{II}}|}$ exactly.

Joint weight — same start offset. Both cuts are conditions on the *same* uniform offset u (comonotone, not independent; a product would double-penalize):

$$w(q, \tilde{D}) = \text{clip} \left(\frac{\min(t_{f,s,\Phi}(q), t_{\text{lim}}(q, \tilde{D})) - t_{p,\text{eff}}(q)}{t_{\text{cad}}}, 0, 1 \right). \quad (25)$$

Because w now depends on $\tilde{D}$, the q -integrand's distance volume is no longer $[\tilde{D}_{\text{eff}}^3 - \tilde{D}_{\text{min}}^3]_+ P$; it becomes the **closed-form weighted volume**

$$V_w(q) = \int_{\tilde{D}_{\text{min}}}^{\tilde{D}_{\text{eff}}(q)} 3\tilde{D}^2 w(q, \tilde{D}) d\tilde{D} = \frac{w_f}{t_{\text{cad}}} \left[\min(\tilde{D}_{\text{eff}}, \tilde{D}_c)^3 - \tilde{D}_{\text{min}}^3 \right]_+ + \frac{t_{p,\text{eff}}}{t_{\text{cad}}} [G(U) - G(L)]_+, \quad (26)$$

with the fade window $w_f = \text{clip}(t_{f,s} - t_{p,\text{eff}}, 0, t_{\text{cad}})$, ramp start $\tilde{D}_c = \tilde{D}_0 [t_{p,\text{eff}} / (t_{p,\text{eff}} + w_f)]^{1/k}$, antiderivative $G(\tilde{D}) = 3\tilde{D}_0^k \tilde{D}^{3-k} / (3-k) - \tilde{D}^3$ (of $3\tilde{D}^2[(\tilde{D}_0/\tilde{D})^k - 1]$), and clamped ramp edges $U = \min(\tilde{D}_{\text{eff}}, \tilde{D}_0)$, $L = \min(\max(\tilde{D}_{\text{min}}, \tilde{D}_c), U)$. The exponent domain is safe: $k_{\text{II}} = 8/[3(p-1)] \leq 2.64$ and $k_{\text{III}} = 2/p < 1$ for the app's $p \in [2.01, 3]$, so the $k = 3$ pole is unreachable. The pointwise w (25) drives the dR/dD view and the numerical D -median; V_w (26) drives the rate integral, the numerical q -median, and the dR/dq view. Numerics: $\eta^{-1/2}$ is computed directly as $10^{-E/2}$ so that extreme $s_{\text{rise}} t_{\text{cad}}$ (where η overflows) degrades gracefully to $\tilde{D}_0 = 0$, $V_w = 0$.

Limits and invariants. (i) $s_{\text{rise}} = 0$ bypasses everything bit-identically (both modes, all derived quantities). (ii) Monotone non-increasing in s_{rise} , and jointly in s_{fade} . (iii) Pointwise $w \leq \min(P_{\text{fade}}, \mathbf{1}\{\tilde{D} \leq \tilde{D}_0\})$, so the joint rate sits below each single-cut rate, and the weighted full-integral rate sits below the best-case ($u = 0$) hard-cut rate built from the same $\tilde{D}_{\text{max}}$ profile. *Note:* full vs dominant themselves are **not** ordered under the rise cut — the dominant truncation drops the D -structured tail beyond the boundary angle, which can outweigh the uniform-start suppression the full mode applies. (iv) $s_{\text{rise}} \rightarrow 0^+$ does *not* limit to the bypass near the flux boundary ($\tilde{D} \rightarrow \tilde{D}_{\text{max}}$): any $s_{\text{rise}} > 0$ also switches on the real “catch it before it fades to the limit” physics that the baseline neglects (sources near their maximum detectable distance are above the limit only briefly). $s_{\text{rise}} = 0$ disables previous-visit modeling entirely; the residual is confined to the near-boundary shell.

Assumptions. (a) The rise itself is ignored (dominant-term lightcurve), which is what makes the previous-visit non-detection automatic. (b) The start offset u is decoupled from the detection-count machinery (q_i , D_i keep their dominant-term forms). (c) t_{lim} uses the single power-law phase of the viewer; a crossing into the post- t_j (steeper) decay makes the true window slightly *smaller* than modeled, so the rise treatment is mildly permissive there — opposite in sign to the fade filter’s conservative single-phase bias.

Implementation reference. `_rise_eta()` returns $(\eta, \eta^{-1/2})$; `_t_p_eff()` the floored peak time; `_rise_fade_windows()` the shared per-point quantities; `_joint_survival()` the pointwise w (25); `_weighted_D_volume()` the closed form (26). The dominant-mode boundary shifts and the on-axis max-term live in `rate_log10` (and, in capped- q_{max} form, `compute_medians_analytic`); the bridge’s dominant $R(q)/R(D)$ view applies the per-regime caps but keeps the separable regime form (no max-term) — use full-integral mode for the deep-cut corner there.

6.7 Legacy off-axis-only mode (deprecated)

Earlier versions of the code exposed an “off-axis detections only” boolean that subtracted the on-axis ($q < q_{\text{dec}}$) contribution: $R_{\text{off}} = R_{\text{total}} - R_{\text{on}}$. This was replaced by the continuous q_{min} filter described above. Setting $q_{\text{min}} \approx q_{\text{dec}} \approx 1.03$ approximates the old behaviour: the angular shell $q < q_{\text{dec}}$ is excluded directly from each regime’s integral, rather than computed and subtracted. The new formulation is numerically cleaner (no near-cancellation of two large numbers) and exposes the filter as a continuous physics parameter rather than a hard toggle.

7 Detection-Window Settings

Two independent, off-by-default settings refine how the “ $\geq i$ detections within the cadence” requirement is counted. Both off reproduces the legacy simple mode bit-for-bit (parity-tested); both on gives the corrected criterion

$$t_+(D) - t_{p,\text{eff}}(q) \geq (i-1) t_{\text{cad}}. \quad (27)$$

7.1 Setting 1: $(i - 1)$ cadence intervals (`win_i_minus_one`)

i detections spaced t_{cad} apart bracket only $i - 1$ gaps, so the required detectability duration is

$$T_{\text{req}} = (i - 1) t_{\text{cad}} \quad \text{instead of} \quad T_{\text{req}} = i t_{\text{cad}},$$

substituted everywhere T_{req} enters (q_i , D_i , and the window-mode D_{eff} below; helper `_t_req_s()`). Exactly equivalent to the legacy criterion evaluated at $i - 1$. Degenerate case $i_{\text{det}} = 1$: $T_{\text{req}} = 0$, so $q_i \rightarrow 1$ and $D_i \rightarrow +\infty$ — a single detection at the peak needs no duration and the cadence constraint vanishes.

7.2 Setting 2: window measured from the peak (`win_from_peak`)

The legacy criterion measures the window from the *burst*: $t_+(D) \geq T_{\text{req}}$, with the boundary pair q_i ($t_p = T_{\text{req}}$) and D_i ($t_+ = T_{\text{req}}$) forming the admissible rectangle $\{q \leq q_i\} \times \{D \leq D_i\}$. But the off-axis flux is negligible before the peak, so epochs in $[0, t_p]$ cannot be detections: at the dominant corner (q_i, D_i) the true above-threshold window $t_+ - t_p$ is ≈ 0 even though i detections were credited. The corrected requirement counts the window from the turn-on $t_{\text{on}} \approx t_p$:

$$t_+(D) \geq t_{p,\text{eff}}(q) + T_{\text{req}}, \quad (28)$$

which turns the cadence cap into the q -dependent distance

$$D_w(q) = D(t_+ = t_{p,\text{eff}}(q) + T_{\text{req}}), \quad D(t_+) = \begin{cases} D_{\text{dec}} (t_+/t_{\text{dec}})^{\alpha_{\text{II}}/2}, & t_+ < t_j, \\ D_{\text{dec}} (t_j/t_{\text{dec}})^{\alpha_{\text{II}}/2} (t_+/t_j)^{\alpha_{\text{III}}/2}, & t_+ \geq t_j, \end{cases} \quad (29)$$

with $\alpha_\Phi < 0$ the temporal flux indices (`D_from_t_plus()`, `D_eff_window_cm()`). The inversion $D(t_+)$ unifies the two legacy rectangle boundaries: at $t_+ = T_{\text{req}}$ it reproduces D_i exactly, and at $t_+ = t_p(q)$ it reproduces the flux-limited $D_{\text{max}}(q)$. Since $t_{p,\text{eff}} + T_{\text{req}} > t_p$, one has $D_w(q) < D_{\text{max}}(q)$ everywhere — the window cap is always the binding distance constraint, and the full-integral integrand becomes $q [\min(\tilde{D}_w(q), 1)]^3$.

Consequences. (i) **No dominant-term closed form:** D_{eff} is q -dependent, and the leading integral $\int q \tilde{D}_w(q)^3 dq$ is elementary only in phase III. `rate_log10` therefore evaluates the q -integral internally under this setting (in both UI modes), `compute_medians` always takes the numerical path, and the bridge's $R(q)/R(D)$ views use the full-integral branch. Region masks keep the legacy q_i boundaries as an approximate classification (regime colouring only). (ii) **Off-axis suppression:** in the phase-III cadence-limited regime the corrected integral evaluates to $1/(3p/2 - 1)$ of the legacy rectangle (≈ 0.43 at $p = 2.2$; on-axis contributions are unchanged since $t_{p,\text{eff}} = t_{\text{dec}} \ll T_{\text{req}}$), and the dominant angle shifts inward from $\tilde{q}_i$ to $\tilde{q}_i/\sqrt{3p - 1}$ — the detected q -median moves on-axis, toward the observed ZTF $q_{\text{med}} \lesssim 1$. (iii) **Limits:** $t_{\text{cad}} \rightarrow 0$ gives $D_w(q) \rightarrow D_{\text{max}}(q)$ and the legacy rate is recovered; the $s_{\text{fade}}/s_{\text{rise}}$ weights compose unchanged (they constrain the first-detection timing, the window constrains the duration).

Implementation reference. Flags on `DetectionRateModel` (constructor flags `win_i_minus_one` and `win_from_peak`, threaded through `make_rate_model`; bridge params `win_iminus1` / `win_tp`; two Settings switches in the app). Tests: `tests/test_window_toggles.py` (parity, the $i - 1$ shift identity, $D(t_+)$ boundary identities, suppression/limit checks).

8 Median Viewing Angle and Distance

The model computes the median viewing angle q_{med} and median detection distance D_{med} of the detected burst population at each grid point. Two computation modes are available, matching the rate mode.

8.1 Analytic medians (dominant-term mode)

In the dominant-term approximation, the effective detection volume is uniform within an angular cap $[0, q_{\max}]$ at constant distance $D_{\text{eff, const}}$. The marginal distributions are:

$$p(q) \propto q, \quad q \in [0, q_{\max}] \quad \implies \quad q_{\text{med}} = \frac{q_{\max}}{\sqrt{2}},$$

$$p(D) \propto D^2, \quad D \in [0, D_{\text{eff, const}}] \quad \implies \quad D_{\text{med}} = D_{\text{eff, const}} \cdot 2^{-1/3}.$$

The quantities $q_{\max}$ and $D_{\text{eff, const}}$ depend on the active regime:

Regime	$q_{\max}$	$D_{\text{eff, const}}$
A1	q_{nr}	D_{euc}
A2, A3	q_{Euc}	D_{euc}
A4	q_{nr}	D_i
A5, A6	q_i	D_i
A7	q_{dec}	D_{dec}

The $p(q) \propto q$ distribution arises from the volume element $\theta_j^2 q dq$ in the solid-angle integral. The $p(D) \propto D^2$ distribution reflects uniform 3-D number density at fixed angular slice.

8.2 Numerical medians (full-integral mode)

When full-integral mode is active, medians are computed from the same $\tilde{D}_{\text{eff}}(q)$ profile used in Eq. (13).

Median q : Define the unnormalized weight $w(q) = q \cdot \tilde{D}_{\text{eff}}(q)^3$. Then q_{med} is found from the cumulative integral:

$$\frac{\int_0^{q_{\text{med}}} w(q) dq}{\int_0^{q_{\text{nr}}} w(q) dq} = \frac{1}{2}. \quad (30)$$

Evaluated by cumulative trapezoidal sum over $N_q = 200$ points.

Median D : Define $Q(D) = \max\{q : \tilde{D}_{\text{eff}}(q) \geq D/D_{\text{euc}}\}$ (the largest angle still detectable at distance D). Since $\tilde{D}_{\text{eff}}(q)$ is non-increasing, $Q(D)$ is obtained by counting how many grid points satisfy the threshold. The marginal density is:

$$p(D) \propto D^2 Q(D)^2, \quad (31)$$

which reduces to D^2 for uniform density and constant Q . The median is found by cumulative summation over a grid of 100 normalized distance levels $D/D_{\text{euc, eff}} \in [0, 1]$, where $D_{\text{euc, eff}} = \tilde{D}_{\text{eff}}(0) \cdot D_{\text{euc}}$.

9 Simplified Luminosity Function

9.1 Definition and physical meaning

At fixed light-curve shape, let the on-axis spectral luminosity at deceleration be

$$L \equiv L_{\nu}(t_{\text{dec}}), \quad \frac{d\mathcal{R}}{dL} = \frac{\mathcal{A}}{L_{\text{ref}}} \left(\frac{L}{L_{\text{ref}}} \right)^{\alpha}, \quad 0 < L < \infty, \quad (32)$$

where $L_{\text{ref}} = 10^{32} \text{ erg s}^{-1} \text{ Hz}^{-1}$ is fixed. The app defaults to $\mathcal{A} = 1 \text{ Gpc}^{-3} \text{ yr}^{-1}$ and $\alpha = -2$. Luminosity is defined at the selected observing band. The normalization has the particularly simple interpretation

$$\left. \frac{d\mathcal{R}}{d \ln L} \right|_{L=L_{\text{ref}}} = \mathcal{A}.$$

Per decimal logarithm the corresponding density is $\ln(10)\mathcal{A}$. This is an event-rate density, not a probability density: no power law can be integrated over all positive luminosities to give a finite intrinsic rate. Consequently there is no intrinsic population median or finite R_{int} for this model. The app's single-luminosity $\mathcal{R}$ control does not set its normalization.

The LF changes only brightness. All bursts share the selected p , timescales, and angular light-curve shape. Write the flux as

$$F_\nu(t, D, q; L) = \frac{L}{4\pi D^2} \tilde{F}_\nu(t, q), \quad \tilde{F}_\nu(t_{\text{dec}}, 0) = 1. \quad (33)$$

The pure amplitude factors ϵ_e , ϵ_B , and ν in PLS G cancel from this normalized shape. They, the flux override, and the single-luminosity $\mathcal{R}$ control are inactive while LF is enabled. Changing p , $E_{k,\text{iso}}$, n_0 , Γ_0 , or θ_j can still change slopes, timescales, or angles. Changing D_{Euc} changes the volume.

9.2 Why the detected rate converges

In the unfiltered Euclidean volume, a sufficiently faint burst has accessible volume proportional to $L^{3/2}$. Hence its contribution per logarithmic luminosity interval is proportional to $L^{\alpha+5/2}$. At sufficiently high luminosity the accessible volume saturates, and that contribution becomes proportional to $L^{\alpha+1}$. Both tails are integrable precisely when

$$-\frac{5}{2} < \alpha < -1. \quad (34)$$

The boundary slopes diverge logarithmically. The app enforces the open mathematical interval; the slider spans -2.49 to -1.01 . The spectral luminosity at the Figure 15 Euclidean corner is

$$L_{\text{dec}} = 4\pi D_{\text{Euc}}^2 F_{\text{lim}}.$$

Cadence requirements raise this threshold through the flux at the required last detection. A maximum near the resulting corner does not imply a narrow population: either luminosity tail becomes very broad near its convergence boundary. The standalone *Simplified Luminosity Function* report gives the complete derivation and contribution distributions.

9.3 Full integral and scale invariance

Let $w(L, D, q)$ be the detection selection weight, including the shared random start of any active rise and fade requirements. The full rate is

$$R = 4\pi f_\Omega \theta_j^2 \int_{q_{\text{min}}}^{q_{\text{nr}}} q dq \int_{D_{\text{min}}}^{D_{\text{Euc}}} D^2 dD \int_0^\infty \frac{d\mathcal{R}}{dL} w(L, D, q) dL. \quad (35)$$

All lengths and the volumetric density in this expression must use the same length unit. In particular, the implementation converts Gpc to cm explicitly. For a hard selection at angle q , let $L_{\text{det}}(q, D_{\text{Euc}})$ denote the minimum spectral luminosity detectable at the Euclidean wall, including the requested cadence or peak-time requirement. At another distance this threshold

is multiplied by $(D/D_{\text{Euc}})^2$. Thus

$$\int_{L_{\text{det}}(q, D_{\text{Euc}})}^{\infty} \frac{d\mathcal{R}}{dL} dL = \frac{\mathcal{A}}{-\alpha - 1} \left[\frac{L_{\text{det}}(q, D_{\text{Euc}})}{L_{\text{ref}}} \right]^{\alpha+1} \left(\frac{D}{D_{\text{Euc}}} \right)^{2\alpha+2}, \quad (36)$$

$$R = \frac{4\pi f_{\Omega} \theta_j^2 \mathcal{A} D_{\text{Euc}}^3}{(-\alpha - 1)(2\alpha + 5)} \left[1 - \left(\frac{D_{\text{min}}}{D_{\text{Euc}}} \right)^{2\alpha+5} \right] \times \int_{q_{\text{min}}}^{q_{\text{nr}}} q \left[\frac{L_{\text{det}}(q, D_{\text{Euc}})}{L_{\text{ref}}} \right]^{\alpha+1} dq. \quad (37)$$

For probabilistic selection, the engine integrates the corresponding constant and power-law pieces of w analytically in luminosity. Rise and fade share the same start time; their separate survival probabilities are not multiplied. Distance integration remains analytic because all selection dependence on luminosity and distance occurs through L/D^2 . The remaining angular integral is evaluated numerically, resolving physical breakpoints. This is the app's Exact mode within the stated light-curve model.

At fixed cadence and shape, with no distance floor,

$$R \propto \mathcal{A} F_{\text{lim}}^{\alpha+1} D_{\text{Euc}}^{2\alpha+5}. \quad (38)$$

With a fixed nonzero distance floor the radial factor is instead $D_{\text{Euc}}^{2\alpha+5} - D_{\text{min}}^{2\alpha+5}$. There are no hidden luminosity cutoffs or finite-luminosity interpolation tables. If one changes the reference convention, the same physical intensity requires

$$\mathcal{A}' = \mathcal{A} \left(\frac{L'_{\text{ref}}}{L_{\text{ref}}} \right)^{\alpha+1}.$$

9.4 Most-contributing-region approximation

With Exact mode off, each luminosity contributes its dominant rectangular region in the (q, D) plane. If its upper bounds are $q_{\star}(L)$ and $D_{\star}(L)$, its contribution is

$$R_{\text{approx}} = \frac{4\pi f_{\Omega} \theta_j^2}{6} \int_0^{\infty} \frac{d\mathcal{R}}{dL} [q_{\star}(L)^2 - q_{\text{min}}^2]_+ [D_{\star}(L)^3 - D_{\text{min}}^3]_+ dL, \quad (39)$$

where $[x]_+ = \max(x, 0)$. Flux, cadence, jet-break, and filter crossings partition luminosity into intervals. On each interval the integrand is a sum of powers of physical luminosity. Each term therefore uses

$$\int_{L_a}^{L_b} L^{\gamma} dL = \begin{cases} (L_b^{\gamma+1} - L_a^{\gamma+1})/(\gamma + 1), & \gamma \neq -1, \\ \ln(L_b/L_a), & \gamma = -1. \end{cases}$$

The faint and bright intervals use the exact limits 0 and ∞ . Numerically stable power differences handle nearly logarithmic finite intervals. Inactive coefficients must not be multiplied by divergent moments. For strong rise cuts, choose the larger integrated angular or on-axis rectangle contribution on each base interval. Retain that choice when querying distributions; a changed plotting threshold does not define a new selected population. Regime coloring identifies the largest integrated regime contribution.

The approximation is an integral of dominant regions, not a replacement of the LF by a single typical luminosity. Its accuracy depends on slope and selection; it need not be close near either convergence boundary. A requested peak-time-dependent horizon follows the existing full-integral fallback.

9.5 Detected distributions and interface behavior

In full mode, for $D_{\min} \leq d \leq D_{\text{Euc}}$,

$$P(D < d) = \frac{d^{2\alpha+5} - D_{\min}^{2\alpha+5}}{D_{\text{Euc}}^{2\alpha+5} - D_{\min}^{2\alpha+5}}, \quad D_{\text{med}} = \left[\frac{D_{\text{Euc}}^{2\alpha+5} + D_{\min}^{2\alpha+5}}{2} \right]^{1/(2\alpha+5)}. \quad (40)$$

With no distance floor this becomes $D_{\text{med}} = D_{\text{Euc}} 2^{-1/(2\alpha+5)}$. The normalized distance distribution is independent of flux sensitivity, cadence, and angle cuts, provided the selected population is nonempty and the model remains homogeneous in L/D^2 .

Approximate-mode distributions integrate intersections of the selected rectangles with the requested angle or distance threshold. Their medians are medians of that mixture, never averages of the component medians. Analytic differentials and logarithmic distance inversion preserve the very small medians possible as $\alpha \rightarrow -5/2$.

All app rates, optimization results, slices, markers, and distributions use the selected calculation mode. The plotted cumulative curves are survival rates $R(q \geq q_{\text{cut}})$ and $R(D \geq D_{\text{cut}})$ within the selected sidebar filters. Their percentage normalization uses the full selected rate, not the first point of a finite plotting grid. An annotation discloses appreciable probability below the displayed range.

The LF switch remains in Settings; α and $\log_{10} \mathcal{A}$ appear in Parameters. Intrinsic counters and the single-burst normalization readout are hidden in LF mode, and inactive controls are disabled. Their stored values are restored when LF is switched off. Finite-cutoff reference calculations remain available only through an explicit Python constructor for scientific validation; they are not an app setting.

10 Optical Cadence Constraints

For optical surveys, only three cadence regions are physically valid:

- **Continuous** ($i \cdot t_{\text{cad}} < t_{\text{night}}$, equivalently $t_{\text{cad}} < t_{\text{night}}/i$): sub-night sampling. The afterglow peak is required to occur within a single night. Because only nighttime observations are possible, the effective sky coverage per day is reduced by the night fraction:

$$\log_{10} R_{\text{eff}} = \log_{10} R_{\text{night}} + \log_{10} \left(\frac{t_{\text{night}}}{1 \text{ day}} \right),$$

where R_{night} is evaluated with $f_{\text{live,night}} = f_{\text{live}}/f_{\text{night}}$, and the $\log_{10}(t_{\text{night}}/\text{day})$ factor accounts for the fact that the survey can only operate during the nighttime fraction $f_{\text{night}} = t_{\text{night}}/86400$.

- **Gap** ($t_{\text{night}}/i \leq t_{\text{cad}} < 1 \text{ day}$): *physically forbidden*. At these cadences i detections within one night are impossible, yet the cadence is sub-daily. Grid points in this band are marked invalid (NaN). *Note:* for $i = 1$ a single detection fits in any night, so the gap region does not apply and would extend only from t_{night} to 1 day; this case is not yet fully handled in the current implementation.
- **Discrete** ($t_{\text{cad}} = n \times 1 \text{ day}$, $n \in \mathbb{Z}^+$): multi-night sampling. Only integer multiples of 1 day are permitted (enforced as a validity mask, not by rounding). No night-fraction penalty applies.

In continuous mode the code uses a separate model instance (`model_night`) with $f_{\text{live,night}} = f_{\text{live}}/f_{\text{night}}$. The $\log_{10}(f_{\text{night}})$ offset is applied post-hoc in log space.